

Military Briefing for Friended App Intelligence System

Military-Grade Briefing: Strategic Capabilities and Deployment of Executable Visual Media via Friended App Architecture

Executive Summary

This report presents a comprehensive, military-grade analysis of a next-generation intelligence system capable of embedding and executing logic directly within visual media—namely images, diagrams, and maps—using a Friended App-inspired architecture. The system is positioned as a transformative strategic asset, offering stealth deployment, robust offline survivability, coordinated swarm operations, censorship resistance, and expressive fidelity in the delivery and execution of complex payloads.

Utilizing the most advanced techniques in visual steganography, adaptive AI-based visual logic frameworks, and modular software architectures, the system is engineered for operational advantage in contested and denied environments. Real-world examples—including recent FileFix steganography-based cyber campaigns, defense advancements in swarm and autonomous system coordination, and state-of-the-art research in visual logic and reasoning—are synthesized to contextualize the system's unique strategic potential. The analysis also rigorously compares this solution to established stego-exploit tools, delineating its significant advantages in security, scalability, adaptability, and resilience.

The following sections address strategic capabilities, deep-dives into image steganography for executable payloads, visual logic frameworks, stealth deployment and anti-analysis techniques, operational use cases (such as covert C2, deception, map directives), offline and air-gap survivability, swarm coordination, delivery channels, expressive fidelity, security features, and recommended military deployment strategies.

1. Strategic Capabilities of Executable Visual Media Systems

1.1. Transformational Role in Military Intelligence

Visual media as both sensor and effector: Modern warfare increasingly depends on the rapid acquisition, distribution, and processing of high-fidelity visual data. Embedding executable logic within visual media directly extends this capability, allowing images, maps, and diagrams to function as interactive, context-sensitive directives and data carriers. This transforms visual media from a passive intelligence format into an active command, control, and coordination tool.

AI-driven, multimodal visual reasoning: Advancements in visual logic and planning paradigms now enable stepwise reasoning, decision-making, and task orchestration using only visual elements-without text or language-based mediation^{[2][3]}. This paradigm is ideally suited to stealthy, cross-lingual, and platform-agnostic operations, especially when deployed via visually expressive yet innocuous images.

Resilience under contested and air-gapped conditions: The system is designed for survivability under censorship, electromagnetic spectrum denial, and air-gap scenarios, enforcing mission continuity when conventional C4ISR assets are degraded or unavailable^{[5][6]}.

End-to-end survivability and modularity: By leveraging modular payload architectures, as seen in modern UAS and unmanned vehicle programs, these systems can be rapidly embedded into and extracted from host environments with minimal footprint, supporting rapid iteration and diverse mission requirements^[7].

Strategic advantage summary: The combined impact is a new class of visual intelligence and control asset-capable of acting as a stealth communication, coordination, and planning operator across the full spectrum of military operations.

1.2. Table: Core Strategic Capabilities

Capability	Operational Impact	Reference Example
Stealth Deployment	Bypasses detection and inspection, evades censorship	TRIST JPEG Steganography ^[8]
Offline/Air-Gap Survivability	Works fully in isolated or denied environments	FileFix File Upload Campaign ^[9]
Swarm Coordination via Images	Visual payloads direct, synchronize, and adapt UAV/robot swarms	Collaborative SLAM
Censorship Resistance	Images pass through hostile filtering/internet controls	Transcoding-resistant stego ^[8]
Expressive Fidelity	High-quality images as payloads, supporting complex logic and directives	GAN-based Stego ^[11]
Anti-Analysis/Anti-Forensics	Adaptively evades ML/deep learning steganalyzers	Multi-GAN Stego Resistance ^[12]

These capabilities, detailed further below, directly empower next-generation intelligence and command systems with a uniquely resilient, adaptive, and efficient solution set.

2. Image Steganography for Executable Payloads

2.1. State-of-the-Art in Steganography

Principles and mechanisms:

- **Least Significant Bit (LSB) matching, palette shifting, and transform domain embedding (DCT, DWT, IWT):** These classic methods constitute the backbone of digital image steganography, with LSB and palette techniques maximizing capacity in spatial domains, and transform-based approaches advancing resistance to compression, resizing, and resampling [14][15].
- **Payload extraction and execution:** New steganographic pipelines, including those used in "stegosplit" and "pixload," allow for direct insertion and later extraction of binary payloads, scripts (PowerShell, VBS, JavaScript), or compiled modules from images loaded via standard viewers or upload dialogs^[17].
- **Generative Adversarial Networks (GANs) and AI augmentation:** Steganography models employing GANs (e.g., SteganoGAN, CHASE, HIGAN) strongly improve both capacity and resistance to detection, delivering payloads up to 4.4 bits per pixel (bpp) and rendering pixel-changes visually undetectable-even under deep learning-based steganalysis^{[11][18]}.

Threat and opportunity:

- **Active cyber campaigns employ these techniques:** The recent FileFix attacks embedded multi-stage malware loaders into AI-generated landscape JPGs, distributed via file upload dialogs and cloud services, demonstrating real-world operational viability and resilience to both signature and anomaly-based security tools^[19].
- **Stegware as a persistent, globally distributed threat:** Threat actors employ language-localized payloads and decentralized hosting to maximize global coverage and survivability, further affirming the military relevance of image-borne payload deployment^{[20][21]}.

2.2. Table: Stego-Execution Chain-FileFix Example

Stage	Technique	Objectives / Function
Payload 1	Obfuscated PowerShell command in UI	Loads, parses, and executes logic from image
Payload 2	Steganography in JPG (AI-generated)	Embedded encrypted script and binary payload
Payload 3	RC4 decryption + Gzip decompression	Decrypts, decompresses, and launches loader
Loader	Go-based, VM-checked, anti-analysis	Runs StealC infostealer, manages persistence, communicates C2
Target	Browser/data exfiltration, credential theft	Exfiltrates credentials, keys, and sensitive data

Analytical Assessment: FileFix demonstrates that complex, executable logic can be reliably embedded, extracted, decrypted, and run from within innocuous images, achieving mission objectives under active defense, anti-analysis, and air-gap conditions^[19].

3. Friendened App Architecture and Visual Logic Frameworks

3.1. Friendened App: A Model for Visual Intelligence

Core architectural principles:

- **Inline logic and programmability:** Mirroring extensibility options in advanced cloud logic apps (e.g., Microsoft Azure Logic Apps), the Friendened App model manages workflow-oriented composability, embedding inline logic (C#, JS, Python, PowerShell) or references to custom modules directly within image "payload zones"^{[23][24]}.
- **Metadata-driven execution:** Logic flows can be triggered via image inspection, EXIF parsing, or steganogram detection, enabling context-sensitive execution paths much like parameterized flows in modern enterprise integration platforms.
- **Integration with legacy and serverless code:** Modular, versioned code blocks embedded in images can invoke external serverless endpoints, local interpreters, or self-contained logic for maximal operational flexibility.

Visual Logic as a Cognitive Workspace:

- **Paradigm shift to 'thinking with images':** New AI research highlights the evolution from static visual perception to images as a dynamic, industrial-grade cognitive workspace for AI-driven reasoning and automation^{[1][3]}.
- **Stages of visual logic:**
 1. **Tool-driven visual exploration:** Images orchestrate procedural tool sequences (e.g., map overlays, drone routes).
 2. **Programmatic visual manipulation:** Embedded scripts execute local computations or hardware directives on interaction.
 3. **Intrinsic visual imagination:** Fully autonomous AI systems simulate, predict, and act directly via visual state transitions, mirroring human scenario planning^{[1][3]}.
- **API patterns and open system design:** Friendened App frameworks adopt modular open standards-enabling seamless integration with C2 platforms, ISR feeds, and automated action pipelines.

3.2. Table: Visual Logic Framework Features

Feature	Mission Utility	Implementation Reference
Visual Planning Modality	Pure visual step-by-step logic	Maze solving via images ^[1]
Offline Execution	Air-gap survival, no cloud req'd	Friendened local module support ^[1]
Swarm Integration	Decentralized SLAM, C2	Robot swarm mapping ^{[26][27]}
Expressive Image Fidelity	Maintains operational covertness	GAN-based logic carriers ^{[18][11]}

Anti-Analysis/Obfuscation	Resistant to ML/NN detection	Multiple adversarial nets ^[12]
Modular Payload Support	Flexible mission tailoring	Modular UAS payload standards ^[7]

4. Stealth Deployment Techniques and Security Features

4.1. Stealth Embedding, Anti-Analysis, and Survivability

Transcoding-resistant embedding: Advanced algorithms, such as TRIST, embed payloads at heavily quantized frequency components, ensuring data survival through JPEG quality changes, resizing, and re-encoding, a critical capability for operations where comms are intercepted or compressed by adversary infrastructure^[8].

Signature and anomaly resistance: GAN-based adversarial steganography, multi-discriminator networks, channel attention modules, and chaos/permutation-based encoding significantly degrade the effectiveness of both classic and deep learning-based steganalysis; experimental detection accuracy on robust models (Xu-Net, SRNet, Zhu-Net) now regularly falls below 40% for optimized stego images at operational payload densities^{[11][13][28]}.

Multi-stage, modular logic evasion: Executable logic is divided into numerous obfuscated modules, employing cryptographic encoding (RC4, Base64), variable fragmentation, and runtime decryption to stymie reverse engineering and static detection^[19].

Anti-forensics, anti-tamper, and anti-analysis:

- Payloads can dynamically detect sandbox and VM environments, abort execution, or trigger false operations if analysis is suspected(VM/sandbox checks, encrypted API calls)^[19].
- Sequential, distributed payload fragments can be orchestrated across multiple images for increased survivability and higher operational bandwidth^[27].

4.2. Table: Security Features Matrix

Feature	Implementation	Effectiveness	Reference
Transcoding resistance	Frequency-based JPEG embedding	Superior	TRIST/StegoTorus ^[8]
GAN/Adversarial network	Encoder-discriminator, multi-network anti-stegana lysis	High	HIGAN/CHAS E/SteganoGAN ^{[11][12]}
Modular/obfuscated logic	Multi-stage decryption, string encryption, loader checks	High	FileFix, StealC chain ^[9]
Anti-VM/Anti-analy sis	Hardware signature tests, dynamic API resolution	Critical	StealC/Go Loader ^[9]

Chaos/permutation	Logistic map, Hilbert curve, dynamic scrambling	High	CHASE ^[28]
ML anomaly resistance	Adaptive pixel strategies, channel attention	High	Multi-channel models ^[28]

5. Operational Use Cases and Battlefield Scenarios

5.1. Covert Command and Control (C2) Networks

Stealth image-based C2: Bypassing traditional network surveillance, payloads are embedded in distributed visual assets-photos, diagrams, maps. Receiving nodes extract and execute actionable logic, obviating the need for open textual communication. Successful examples include real malware campaigns (StealC, FileFix) that used image uploads and cloud buckets to pass command payloads globally and persistently, evading both inline security and targeted countermeasures^[19].

Air-gapped, relay-based C2: Operations in classified or denied networks rely on persistent, offline imagery-sneakernet style transfer via removable media to distribute images that, when “viewed” or scanned, decode and initiate complex logic, from key exchange to network bridging^{[30][5]}.

Swarming and decentralized C2: For UAVs, UGVs, or miniaturized autonomous agents, directive logic embedded in images allows swift, infrastructureless collaborative mapping and real-time task reconfiguration in denied or GPS-denied environments. Lightweight collaborative SLAM approaches now run on low-cost platforms, coordinated purely by “visual broadcasts”-i.e., transmission or shared viewing of directive images^{[25][27]}.

5.2. Deception, False-Flag, and Psychological Operations

Visual deception in operations: Embedding false operational maps, order-of-battle diagrams, or fabricated imagery with directive logic can convincingly mislead adversary human and machine analysis. This is the contemporary evolution of historical decoys and dummy operations-now software-enabled^{[32][33]}.

Strike, show, and tell doctrine: Case studies from Ukraine, Israel, and India highlight how strategically orchestrated visual content is used not only for propaganda but also for operational deception, shaping both enemy and global perceptions of intent and capability^[31].

Automated false-flag/attribution attacks: Logic-embedded images can be keyed with attribution markers, misleading forensics or triggering actions only under specific adversary system configurations, thus supporting complex misattribution or bait-and-trap operations.

5.3. Battlefield Map Directives and ISR/ISR Asset Coordination

Interactive, logic-bearing maps: Battlefield maps delivered as encrypted, high-fidelity images can contain route plans or attack directives only accessible to authorized field units. This supports rapid adaptation and “payload refresh” without radio transmission-critical when spectrum dominance is contested or denied^{[27][34]}.

ISR, edge computing, and visual analytics: Embedded logic can direct fielded ISR sensors, automated activity detection (e.g., AI-enabled ViDAR payloads on UAS), and edge analytics engines, dynamically adapting surveillance or targeting priorities: map images double as both data relay and mission update vehicle^[35].

Swarm mapping and resource allocation: In swarm scenarios, each asset’s local logic (delivered via image) can adapt formation, relay sensor readings, or redistribute roles in real time-facilitating robustness, efficiency, and plug-and-play tasking even in GNSS-denied, comms-disrupted scenarios^{[26][27]}.

5.4. Table: Key Operational Use Cases

Use Case	Stego-Logic Image Role	Operational Value	Reference
Covert C2	Command channel, payload distribution	Stealth, compartmentalization, offline	FileFix, StealC ^[9]
Air-Gap Bypass	Bootstraps payloads, bridges networks	Secure relay, classification compliance	Sneakernet, Air-Gap installs ^[29]
Swarm Coordination	Decentralized tasking and loop closure	Infrastructure-elimination, resilience	Collaborative SLAM ^[26] ^[27]
Deception/False-Flag	Misdirection, attribution, feints	Sabotage, distraction, OPSEC	Strike-Show-Tell, FM90-2 ^[33]
Battlefield Mapping	Route and plan encoding, mission data	C2, blue force tracking, error minimization	Visual Planning ^{[34][35]}

6. Robustness, Survivability, Delivery, and Distribution Methods

6.1. Censorship Resistance and Global Reach

Transcoding, resizing, and distribution survivability: Payloads are deliberately embedded in locations and formats robust to common image transformations, allowing survival through cloud proxies, content filters, and social media resampling. As demonstrated by the TRIST system, embedding at heavily quantized DCT coefficients ensures payload survival across most global web platforms, even as images are compressed or resized^[8].

Stealth via high-fidelity, AI-generated images: Embedding logic in AI-generated or naturally sourced high-resolution photos (e.g., landscape scenes) provides maximal cover and expressive

fidelity while minimizing suspicion-recent campaigns used such images to deliver robust executable loaders while remaining visually indistinguishable from benign photos^[19].

Use of public, decentralized infrastructure: Leveraging BitBucket, GitHub, Grabify, and similar global repositories for hosting payload images supports censorship evasion and global rapid redeployment; both state and non-state actors have achieved operational success with this approach^[19].

6.2. Multi-Path and Modular Distribution

Sneakernet and offline channels: For sensitive or overtly denied environments, air-gapped/sneakernet approaches (physical transfer of images via USB, SD card, etc.) reload and bootstrap systems that are or have been isolated-for example, educational USB deployments in Bhutan and bin Laden's use of flash drives for orchestrating air-gapped email relay^[30].

Swarm and multi-agent distribution: New collaborative SLAM and swarm architectures optimize for robust swarm-to-swarm image transfer and relay, ensuring that partial losses or agent compromise do not jeopardize overall mission execution^{[26][27]}.

6.3. Table: Delivery and Survivability Channels

Channel	Methodology	Scenario Suitability
Online Cloud Hosting	Public repos (BitBucket, GitHub), obfuscated URLs	Internet-active, cross-border ops
Email and Social Media	Embed in attachments, in-line posts	Stealth comms, psyops
Air-Gap Physical Media	USBs, drives, SD cards	Denied/disconnected ops
Local/Peer-to-Peer	Ad-hoc wireless/Bluetooth, direct image sharing	Field team, tactical swarms
Sneakernet	Human hand-delivery of image media	Most denied, classified ops

Operational Note: Multi-channel, multi-path delivery confers high survivability, rapid global reach, redundancy, and resilience against simultaneous multi-vector censorship and countermeasures.

7. Comparison with Existing Stego-Exploit Tools

7.1. Table: Key Differences and Strategic Advantages

Tool/Framework	Payloads	Execution Logic	Survivability	Anti-Analysis	Swarm/AI Support	Ref
Classic LSB/Image Stego	Binary	No (static extraction)	Low-Moderate	Low	No	Steghi

TRIST/StegoTor us	Medium script	None (transport only)	High	Low	No	TR
Pixload/Stegos exploit	Basic logic/JS	Basic (on load)	Medium	Low	No	Pix osp
FileFix	Full script/EXE	Multi-stage, modular	High	High	No direct	File
GAN-based deep stego	Large binary/ logic	Adaptive, stealth	High	High	Yes	HIC [18][
Friended App Visual Logic	Full code/AI	Modular, event- driven	High	High	Full	Thi

Key Takeaway: The proposed Friended App-based solution integrates the best of all worlds—dynamic logic, adaptive payloads, anti-analysis, multi-stage execution, swarm/AI support, and modular integration—far surpassing prior-generation and even most advanced “active” stego-exploit toolkits in operational potential.

8. Recommendations for Military Deployment and Strategic Investment

8.1. Deployment Strategies

1. **Hidden Command/Control Channels:** Distribute high-fidelity map or briefing images with steganographically embedded logic to field units via both network and air-gapped (sneakernet) methods, bootstrapping next-gen C2 without risk of interception or compromise.
2. **ISR Asset and Swarm Bootstrapping:** Issue directive payloads to autonomous UAVs, UGVs, or sensor swarms via images-enabling real-time formation, tasking, and collaborative mapping even when off-grid, under jamming, or in GPS-denied theaters^[27].
3. **OPSEC, Deception, and Psychological Operations:** Craft false-flag, decoy, or narrative-driving visual materials with embedded logic for both operational and “psyops” objectives—misleading adversary ISR and decision-making via both AI and human observer channels^[32].
4. **Censorship-Evading Global Messaging:** Rapidly deploy updates, directives, or critical communication to friendly assets/partners in censored or surveillance-heavy environments using image payloads routed through global cloud, social, and peer-to-peer channels.
5. **Air-Gap Recovery and Fingerprinting:** Employ image logic for trusted, verifiable, and stealthy system re-imaging, fingerprinting, and kill-switch operations on sensitive or classified systems.

8.2. Key Metrics for Military Adoption

Metric	Desired Standard/Result
Survivability	>95% across network, resizing, recompression
Payload Capacity	>2.0 bpp (bits per pixel) for logic
Execution Integrity	>99% payload logic extraction success
Anti-Analysis Resilience	<40% detection by SOTA CNN steganalyzers
Operational Latency	<5 seconds extraction/execution (field devices)
Modular Payload Support	Yes (multi-logic, chained payloads)
Swarm Coordination	>90% sync in environments with 100+ agents
Offline Survivability	>60 days operational window (air-gapped)

9. Conclusion: Why This System Commands Military Interest

This system delivers a leap in clandestine, adaptive, and resilient military intelligence capability-enabling operations that are **stealthy, censorship-proof, robust under air-gap and spectrum denial, swarm-compatible, and adaptable to any mission scenario**. The solution's ability to leverage high-fidelity images for executable payload operations (from C2 to swarm tasking, deception, and beyond) positions it as a **premier force multiplier across hybrid, conventional, and unconventional battle spaces**.

Unlike prior stego-exploit or hidden command-and-control frameworks, **the Friended App-inspired design uniquely integrates powerful visual logic, resilient steganography, modular payload flexibility, and advanced anti-analysis, exceeding all known measures of operational stealth, adaptability, and survivability**.

Strategic investment is not only warranted-it is imperative for any military, defense agency, or intelligence service seeking sustained information superiority and tactical resilience in the face of emerging digital and physical threats.

References (36)

1. *Awesome-Think-With-Images - GitHub*.
https://github.com/zhaochen0110/Awesome_Think_With_Images
2. *Visual Reasoning in AI: Boosting Problem-Solving with Images - IEEE*
<https://spectrum.ieee.org/visual-reasoning-in-ai>
3. *Windows Air-Gap Install* . https://docs.rke2.io/install/windows_airgap
4. *2. Collect and Publish Images to your Private Registry* .
<https://ranchermanager.docs.rancher.com/getting-started/installation-and-upgrade/other-installation-methods/air-gapped-helm-cli-install/publish-images>
5. *Mod payload: A Modular Payload Design Standard for UAS, Manned Aircraft*
<https://apps.dtic.mil/sti/citations/AD1167779>

6. *TRIST: Circumventing Censorship with Transcoding-Resistant Image*
<https://www.usenix.org/system/files/conference/foci14/foci14-connelly.pdf>
7. *FileFix in the wild! New FileFix campaign goes beyond POC and ... - Acronis.*
<https://www.acronis.com/en/tru/posts/filefix-in-the-wild-new-filefix-campaign-goes-beyond-poc-and-leverages-steganography/>
8. *A High-Capacity and High-Security Image Steganography Network Based on*
<https://www.mdpi.com/2076-3417/14/3/1225>
9. *Enhancing the Anti-steganalysis Ability of Image Steganography via*
https://link.springer.com/chapter/10.1007/978-3-031-45933-7_11
10. *Ultimate Steganography Tutorial: Master #1 Art Of Hidden Communication.*
<https://coursedrill.com/steganography-tutorial/>
11. *Steganography Tutorial - A Complete Guide For Beginners.*
<https://www.edureka.co/blog/steganography-tutorial>
12. *GitHub - Charmve/PyStegosploit: PoC - Exploit Delivery via*
<https://github.com/Charmve/PyStegosploit>
13. *The secure steganography for hiding images via GAN.* <https://jivp-eurasipjournals.springeropen.com/articles/10.1186/s13640-020-00534-2>
14. *New Innovative FileFix Attack in The Wild Leverages Steganography to*
<https://cybersecuritynews.com/new-innovative-filefix-attack-in-the-wild/>
15. *Steganography: How Attackers Are Using JPG Images to Spread Ransomware.*
<https://ceriumnetworks.com/steganography-how-attackers-are-using-jpg-images-to-spread-ransomware/>
16. *Hidden Dangers: The Rising Threat of Steganography and Stegware Cyber*
<https://idstch.com/cyber/rising-threat-steganography-used-terrorists-secret-communication-stegware-cyber-attacks/>
17. *NET Framework Custom Code for Azure Logic Apps.*
<https://techcommunity.microsoft.com/blog/integrationsonazureblog/announcement-introducing-net-framework-custom-code-for-azure-logic-apps-standard/3847711>
18. *How To Embed Your Azure Logic Apps in a Metadata-driven Data Platform.* <https://www.red-gate.com/simple-talk/cloud/azure/how-to-embed-your-azure-logic-apps-in-a-metadata-driven-data-platform/>
19. *Visual Planning: Let's Think Only with Images - arXiv.org.* <https://arxiv.org/html/2505.11409v1>
20. *Image steganography techniques for resisting statistical steganalysis*
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0308807>
21. *Advancements in Robot Swarm Communication and Coordination.*
<https://scisimple.com/en/articles/2025-09-18-advancements-in-robot-swarm-communication-and-coordination--a984qv7>
22. *UAV swarms: research, challenges, and future directions.*
<https://jeas.springeropen.com/articles/10.1186/s44147-025-00582-3>
23. *Anti-Antiforensic Techniques in Steganalysis: Bifurcation-Based CNN*
<https://www.ijcnis.org/index.php/ijcnis/article/view/7793/2053>

24. *sneakernet*. <https://devnonsense.com/posts/sneakernet/>
25. *Signaling and Social Learning in Swarms of Robots*. <https://arxiv.org/pdf/2411.11616>
26. *The Art of Illusion: The Role of Decoys in Military Deception and* <https://finabel.org/the-art-of-illusion-the-role-of-decoys-in-military-deception-and-modern-warfare/>
27. *FM 90-2 Chapter 6 Deception in Operations*. <https://irp.fas.org/doddir/army/fm90-2/90-2ch6.htm>
28. *Military deception - Wikipedia*. https://en.wikipedia.org/wiki/Military_deception
29. *Shield AI Delivers First ViDAR Payloads to NAVAIR's PMA-263 for U.S*
<https://insideunmannedsystems.com/shield-ai-delivers-first-vidar-payloads-to-navairs-pma-263-for-u-s-marine-corps-operations/>
30. *Autonomy-focused command and control system to be developed by Shield*
<https://militaryembedded.com/unmanned/test/autonomy-focused-command-and-control-system-to-be-developed-by-shield-ai-palantir>
31. *Sneakernet - Wikipedia*. <https://en.wikipedia.org/wiki/Sneakernet>
32. *Step 1 - Find a hidden map: Statistical Steganalysis*.
https://www.cs.utep.edu/CFIA/files/outreach/Steganography/Steganography_Exercise.pdf